GIU Germany Connect — Combined Relational Schema

PK = Primary Key FK = Foreign Key

1. People and Academic

Relational schema/People_Academic

PK = primary key

FK = foreign key

Person (ID[PK], email, phone, account_status, full_name)

Student (ID[PK, FK references Person(ID)], current_semester, GPA, language_level, personal_interests, CV, GIU_student_number, program_id[FK references Program(ID), NOT NULL])

Partner_Professor (ID[PK, FK references Person(ID)], title, partner_department_id[FK references PartnerDepartment(ID), NOT NULL])

Company_Representative (ID[PK, FK references Person(ID)], company_name, sector, position, location)

GIU_Academic (ID[PK, FK references Person(ID)], academic_rank, department_id[FK references Department(ID), NOT NULL])

Program (ID[PK], name, degree_level, total_credit_hours, description, department_id[FK references Department(ID), NOT NULL])

Course (ID[PK], course_code, title, credit_hours, level, program_id[FK references Program(ID), NOT NULL])

Skill (ID[PK], name, category, description)

Faculty (ID[PK], name, description)

Department (ID[PK], name, description, faculty_id[FK references Faculty(ID), NOT NULL])

PartnerDepartment (ID[PK]) — external reference defined in File 2

ResearchArea (ID[PK], name, description, field_id[FK references Field(ID), NOT NULL])

Field (ID[PK], name, description)

Completed (student_id[PK, FK references Student(ID)], course_id[PK, FK references Course(ID)])

Possesses (student_id[PK, FK references Student(ID)], skill_id[PK, FK references Skill(ID)])

Requires (course_id[PK, FK references Course(ID)], prerequisite_course_id[PK, FK references Course(ID)])

Classified_By (program_id[PK, FK references Program(ID)], field_id[PK, FK references Field(ID)])

Interested_In_Partner (partner_id[PK, FK references Partner_Professor(ID)], research_area_id[PK, FK references ResearchArea(ID)])

Interested_In_Academic (giu_id[PK, FK references GIU_Academic(ID)], research_area_id[PK, FK references ResearchArea(ID)])

Constraints: Person specialization is disjoint (D). Each Program requires at least one Field; each GIU_Academic and Partner_Professor requires at least one ResearchArea as shown in the EERD.

2. University and Partnership

University and partner structure

University(universityID[PK], name, country, city, description)

GIU(universityID[PK, FK references University(universityID)])

PartnerUniversity(universityID[PK, FK references University(universityID)], partnershipStatus, contactInformation)

PartnerDepartment(ID[PK], name, description, partnerUniversityID[FK references PartnerUniversity(universityID), NOT NULL])

PartnerProgram(partnerProgramID[PK], name, degreeLevel, description, departmentID[FK references PartnerDepartment(ID), NOT NULL])

Cooperation agreements

CooperationAgreement(agreementID[PK], title, startDate, endDate, file, status, scope, description, giuUniversityID[FK references GIU(universityID), NOT NULL], partnerUniversityID[FK references PartnerUniversity(universityID), NOT NULL], coordinatorAcademicID[FK references GIU_Academic(ID), NOT NULL])

MobilityAgreement(agreementID[PK, FK references CooperationAgreement(agreementID)], seatQuota, exchange, semesters, tuition, doubleDegree)

ResearchAgreement(agreementID[PK, FK references CooperationAgreement(agreementID)], intellectualProperty, fundingFramework)

GeneralMoU(agreementID[PK, FK references CooperationAgreement(agreementID)], intent, collaboration, renewalTerms)

Agreement constraints

University specialization into GIU and PartnerUniversity is disjoint (D). CooperationAgreement specialization is overlapping (O).

Professor visits and relationship relations

ProfessorVisit(visitID[PK], status, followUpReport, accommodation, travel, purpose, startDate, endDate, partnerProfessorID[FK references Partner_Professor(ID), NOT NULL], hostFacultyID[FK references Faculty(ID), NOT NULL], coordinatorAcademicID[FK references GIU_Academic(ID), NOT NULL])

AgendaItem(visitID[PK, FK references ProfessorVisit(visitID)], seqNumber[PK], intendedAudience, activityType, title, date, time, location)

Many to many relationships

PartnerProgramField(partnerProgramID[PK, FK references PartnerProgram(partnerProgramID)], fieldID[PK, FK references Field(ID)])

MobilityEligibleProgram(agreementID[PK, FK references MobilityAgreement(agreementID)], programID[PK, FK references Program(ID)])

ResearchAgreementArea(agreementID[PK, FK references ResearchAgreement(agreementID)], researchAreaID[PK, FK references ResearchArea(ID)])

3. Opportunities and Applications

CORE SUPERCLASS & SUBCLASS RELATIONS (OPPORTUNITY SPECIALIZATION) Opportunity (

opportunityID [PK],

title,

description,

type,

application_start_date,

application_end_date,

capacity,

location_mode,

relevant_dates,

language_requirements,

status

) ThesisTopic (

opportunityID [PK, FK references Opportunity(opportunityID)],

funding_info,

expected_outcomes,

travel_info,
accommodation_info,
academicID [FK references GIU_Academic(ID), NOT NULL],
partnerProfessorID [FK references Partner_Professor(ID), NULLABLE],
submitted_by_personID [FK references Person(ID), NOT NULL]
) ShortProgram (
opportunityID [PK, FK references Opportunity(opportunityID)],
duration,
location,
hosting_arrangements,
universityID [FK references University(universityID), NOT NULL]
) MobilityProgram (
opportunityID [PK, FK references Opportunity(opportunityID)],
mobility_type,
duration,
host_universityID [FK references University(universityID), NOT NULL],
sending_universityID [FK references University(universityID), NOT NULL]
) APPLICATION & WORKFLOW RELATIONS Application (
applicationID [PK],
application_date,
status,
decision_date,
additional_info,
opportunityID [FK references Opportunity(opportunityID), NOT NULL],
studentID [FK references Student(ID), NOT NULL]
) ApplicationDocument (
document_type [Partial Key],
applicationID [FK references Application(applicationID), NOT NULL],
upload_date,
file_reference,
PRIMARY KEY (applicationID, document_type)
) Interview (
interviewID [PK],
interview_date,
time,
location_or_mode,
outcome,
notes,
applicationID [FK references Application(applicationID), NOT NULL]
) Interviewer (
interviewerID [PK],
role_or_position,
personID [FK references Person(ID), UNIQUE, NOT NULL]
) ASSOCIATIVE & MANY-TO-MANY RELATIONSHIP TABLES InterviewInterviewer (
interviewID [PK, FK references Interview(interviewID)],
interviewerID [PK, FK references Interviewer(interviewerID)]
) OpportunityField (

opportunityID [PK, FK references Opportunity(opportunityID)],
fieldID [PK, FK references Field(ID)]
) ThesisTopicResearchArea (
opportunityID [PK, FK references ThesisTopic(opportunityID)],
researchAreaID [PK, FK references ResearchArea(ID)]
) OpportunityRequiredCourse (
opportunityID [PK, FK references Opportunity(opportunityID)],
courseID [PK, FK references Course(ID)]
) OpportunityRequiredSkill (
opportunityID [PK, FK references Opportunity(opportunityID)],
skillID [PK, FK references Skill(ID)]
) OpportunityEligibleProgram (
opportunityID [PK, FK references Opportunity(opportunityID)],
programID [PK, FK references Program(ID)]
)
EERD constraints: Opportunity specialization is disjoint (D) and partial; Opportunity.type must match its subtype (if any).
Each Opportunity requires at least one Field; each ThesisTopic needs at least one ResearchArea.
These rules need enforcement beyond PK/FK constraints.

4. Participation and Research

Relational schema/Participation
PK = primary key
FK = foreign key
Participation (ID[PK], overall_status, start_date, end_date, application_id[FK references Application(applicationID), NOT NULL])
Mobility (ID[PK, FK references Participation(ID)], status_before, status_during, status_after, final_outcome, university_id[FK references University(universityID), NOT NULL])
Program_Participation (ID[PK, FK references Participation(ID)], payment_status, visa_status, travel_status)
Course_Equivalency (mobility_id[PK, FK references Mobility(ID)], host_course_title[PK], course_id[FK references Course(ID), NOT NULL], approval_status, transferred_credits, host_course_credits)
Course (ID[PK]) – external reference defined in File 1
University (universityID[PK]) – external reference defined in File 2
Application (applicationID[PK]) – external reference defined in File 3
Research_Project (ID[PK], title, description, status, start_date, end_date, key_deadlines, agreement_id[FK references ResearchAgreement(agreementID), NULLABLE])
ResearchArea (ID[PK]) – external reference defined in File 1
Classified_By (project_id[PK, FK references Research_Project(ID)], area_id[PK, FK references ResearchArea(ID)])
ResearchAgreement (agreementID[PK]) – external reference defined in File 2
Research_Output (ID[PK], output_type, title, date, reference, status, project_id[FK references Research_Project(ID), NOT NULL])
Funding_Opportunity (ID[PK], name, funding_agency, type, amount, application_deadline, eligibility_conditions, status)
Targets (project_id[PK, FK references Research_Project(ID)], funding_id[PK, FK references Funding_Opportunity(ID)])

Supports (funding_id[PK, FK references Funding_Opportunity(ID)], mobility_id[PK, FK references Mobility(ID)])

GIU_Academic (ID[PK]) — external reference defined in File 1

GIU_Participates_In (giu_id[PK, FK references GIU_Academic(ID)], project_id[PK, FK references Research_Project(ID)], role)

Partner_Professor (ID[PK]) — external reference defined in File 1

Partner_Participates_In (partner_id[PK, FK references Partner_Professor(ID)], project_id[PK, FK references Research_Project(ID)], role)

Notification (ID[PK], message_content, timestamp, category, urgency, read_status, person_id[FK references Person(ID), NOT NULL])

Constraints: Participation specialization into Mobility and Program_Participation is disjoint (D). Each Research_Project requires at least one ResearchArea.